AGILE PRINCIPLES & LEAN PRODUCTION

COMP6100001 - SOFTWARE ENGINEERING
SESSION 03

UNIVERSITAS **BINA NUSANTARA**

SUBJECT MATTER EXPERT

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SESSION 03 – AGILE PRINCIPLES AND LEAN PRODUCTION

LEARNING OBJECTIVES

At the end of this session, students will be expected to be able to:

- Explain the motivation behind Agile & Lean
- Describe Agile Manifesto values & principles
- Explain Lean & Kaizen for continuous improvement
- Analyze how Agile reduces the cost of change

SESSION 03 – AGILE PRINCIPLES AND LEAN PRODUCTION

CONTENT OUTLINE

- Agile principles
- Lean foundations
- Agile Manifesto
- Kaizen / continuous improvement
- Cost of change
- Agile mindset (pre-Scrum)

WHAT IS AGILE?

- The ability to **create and respond to change**
- A way to **succeed in uncertainty**
- Focused on **adaptiveness**, not predictability

WHAT MAKES AGILE DIFFERENT?

- Focus on **people and collaboration**
- Teams are **cross-functional**
- Skills matter more than rigid roles
- Practices support **coordination and transparency**

AGILE IS A MINDSET

- Guided by the **Agile Manifesto's values and principles**
- Helps teams **navigate uncertainty**
- Work in short cycles: **try → get feedback → adapt**

WHERE DID AGILE IDEAS COME FROM?

LEAN PRODUCTION

TOYOTA PRODUCTION SYSTEM (TPS)

Late 1940s

- Build **only what is needed**, when it is needed (pull)
- **Stop and fix problems early** (quality first)
- Teams are **empowered and multi-skilled**
- **Continuous improvement** through small changes (Kaizen)

Lean focuses on learning fast and reducing waste

LEAN PRODUCTION

FROM LEAN TO AGILE

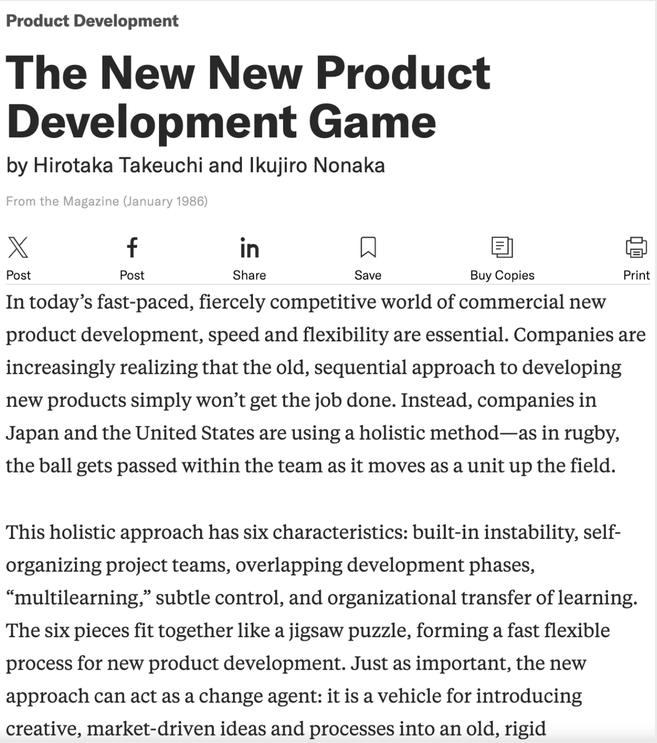
Lean Thinking	Agile Software Development
Pull system	Deliver in small increments
Stop to fix defects	Continuous testing & feedback
Empowered workers	Self-organizing teams
Continuous improvement	Retrospectives & adaptation

Agile applies Lean thinking to software development

AGILE MANIFESTO

AGILE MANIFESTO AND PRINCIPLES

EVOLUTION TO AGILE MANIFESTO



Product Development

The New New Product Development Game

by Hirotaka Takeuchi and Ikujiro Nonaka

From the Magazine (January 1986)

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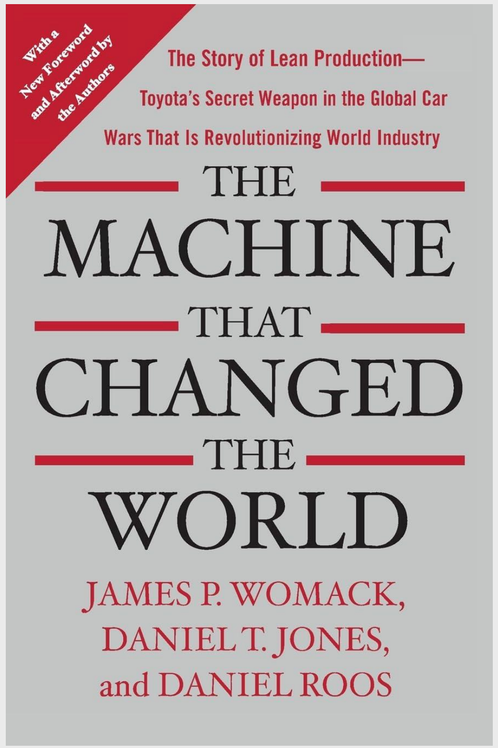
In today's fast-paced, fiercely competitive world of commercial new product development, speed and flexibility are essential. Companies are increasingly realizing that the old, sequential approach to developing new products simply won't get the job done. Instead, companies in Japan and the United States are using a holistic method—as in rugby, the ball gets passed within the team as it moves as a unit up the field.

This holistic approach has six characteristics: built-in instability, self-organizing project teams, overlapping development phases, "multilearning," subtle control, and organizational transfer of learning. The six pieces fit together like a jigsaw puzzle, forming a fast flexible process for new product development. Just as important, the new approach can act as a change agent: it is a vehicle for introducing creative, market-driven ideas and processes into an old, rigid

1986

https://hbr.org/1986/01/the-new-new-product-development-game



1990

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2001

AGILE MANIFESTO

12 PRINCIPLES BEHIND THE AGILE MANIFESTO

1. Highest priority is to **satisfy** the **customer** through **early** and **continuous delivery** of valuable software.
2. Welcome **changing requirements**, even late in development. Agile processes harness change for the customer's **competitive advantage**.
3. **Deliver** working software **frequently**, from a couple of weeks to a couple of months, with a preference to the **shorter** timescale.
4. **Businesspeople** and **developers** must **work together daily** throughout the project.
5. **Build** projects **around motivated individuals**. Give them the environment and support they need, and trust them to get the job done.

AGILE MANIFESTO

12 PRINCIPLES BEHIND THE AGILE MANIFESTO

6. The most **efficient** and **effective method** of conveying information to and within a development team is **face-to-face conversation**.
7. **Working software** is the **primary measure** of **progress**.
8. Agile processes **promote sustainable development**. The sponsors, developers, and users should be able to maintain a constant pace indefinitely.
9. **Continuous attention** to **technical** excellence and good **design** enhances agility.
10. **Simplicity**, the art of maximizing the amount of work not done, is **essential**.
11. The **best architectures**, **requirements**, and **designs** emerge from **self-organizing teams**.
12. At **regular intervals**, the team **reflects** on how to become **more effective**, then tunes and adjusts its behavior accordingly.

AGILE DEVELOPMENT IS *ADAPTIVE* RATHER THAN PREDICTIVE; *PEOPLE-ORIENTED* RATHER THAN PROCESS-ORIENTED. —MARTIN FOWLER

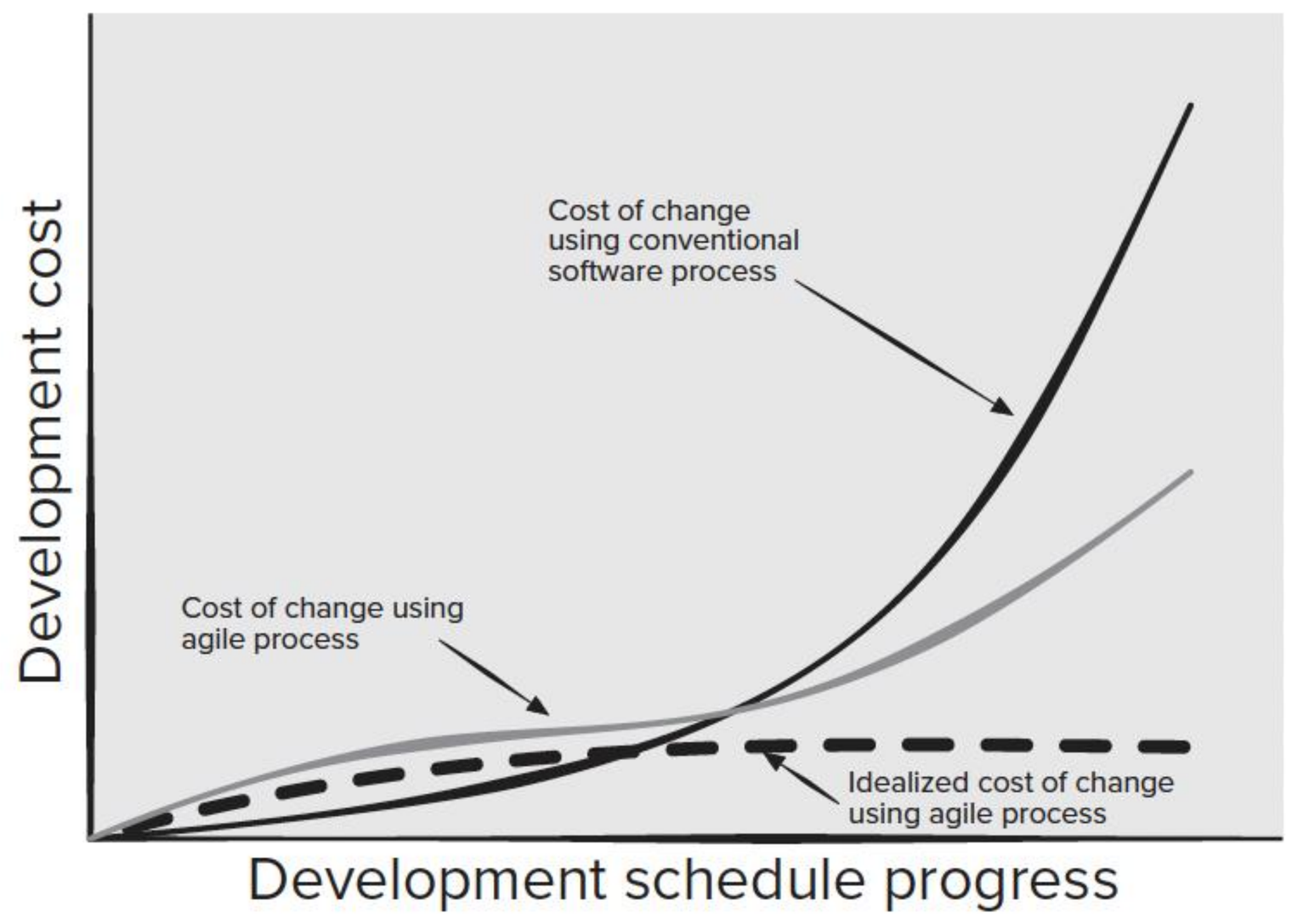
AGILE METHODOLOGY

KAIZEN CYCLE

FOR CONTINUOUS IMPROVEMENT

THE PROCESS

AGILITY AND THE COST OF CHANGE



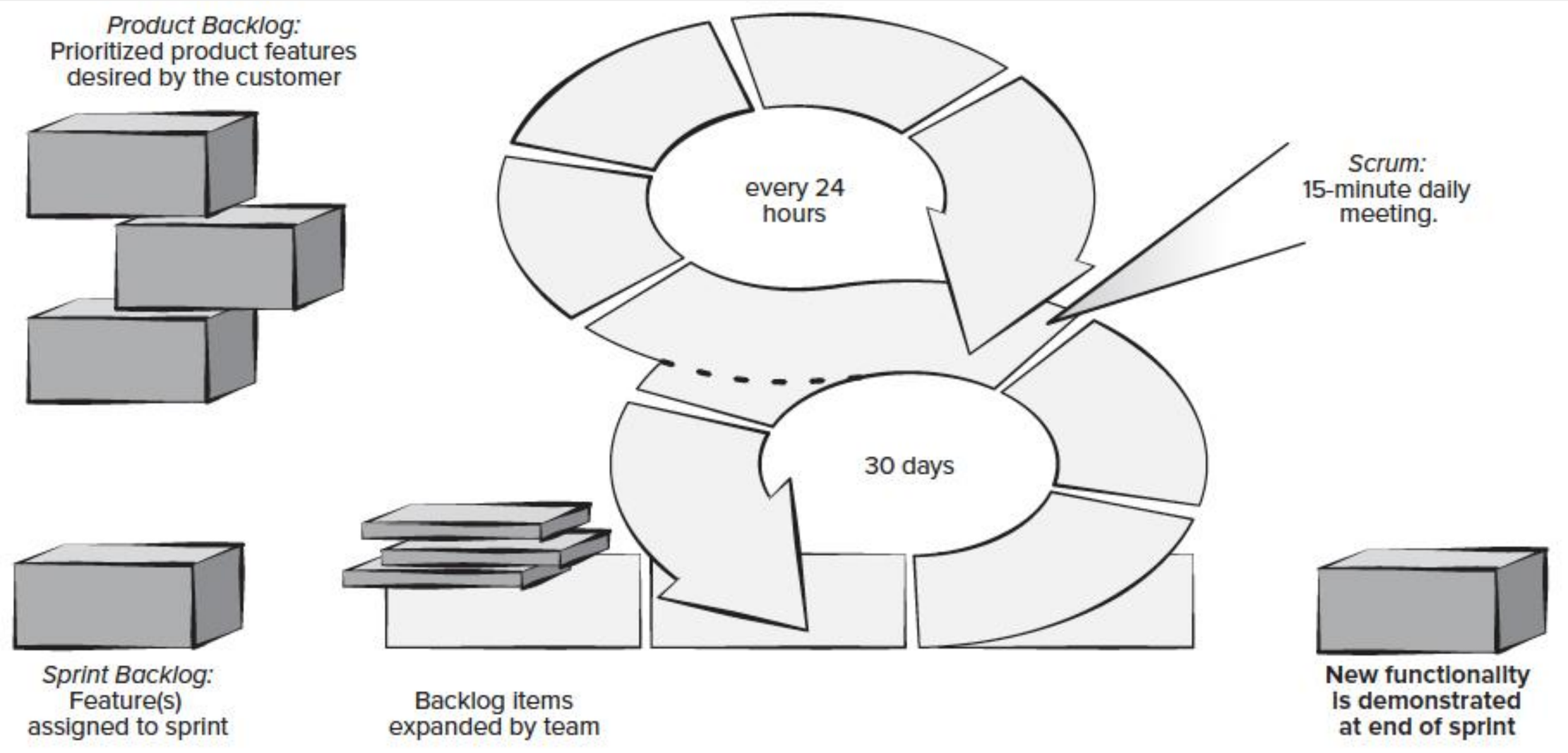
AGILE FRAMEWORK: SCRUM

SCRUM

- Originally proposed by Schwaber and Beedle
- Scrum—distinguishing features
  - Development work is partitioned into “packets”
  - Testing and documentation are on-going as the product is constructed
  - Work occurs in **“sprints”** and is derived from a **“backlog”** of existing requirements
  - Meetings are **very short** and sometimes conducted without chairs
  - **“demos”** are delivered to the customer with the time-box allocated

AGILE FRAMEWORK: SCRUM

SCRUM



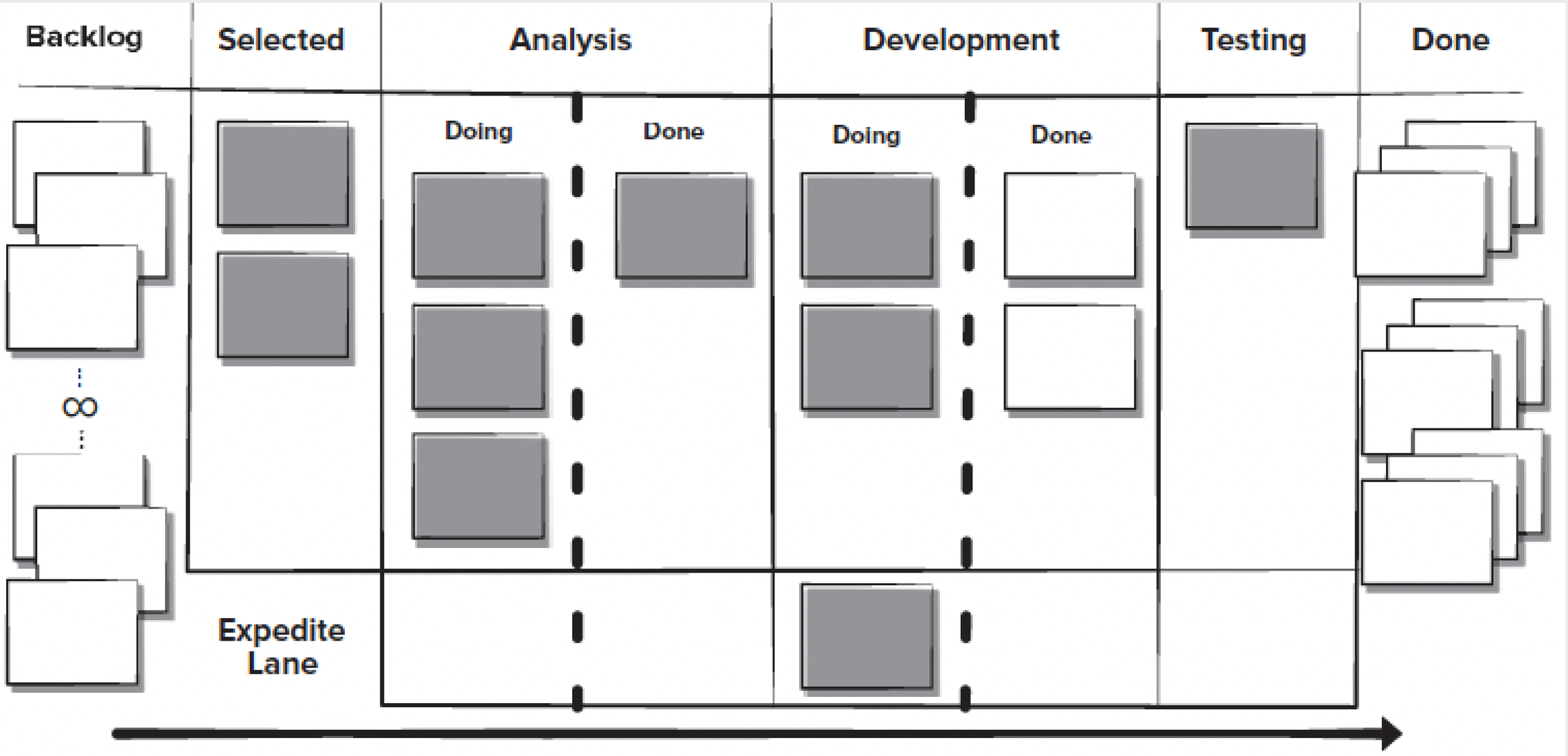
AGILE FRAMEWORKS: KANBAN

DESCRIPTION

- The Kanban method is a lean methodology that describes methods for improving any process or workflow, focused on change management and service delivery
- 6 core practices:
  - **Visualizing workflow using a Kanban board**
    - The Kanban board is organized into columns representing the development stage for each element of software functionality. The cards on the board might contain single user stories or recently discovered defects on sticky notes and the team would advance them from “to do,” to “doing,” and “done” as the project progresses.
  - **Limiting the amount of work in progress (WIP) at any given time**
    - Developers are encouraged to complete their current task before starting another. This reduces lead time, improves work quality, and increases the team’s ability to deliver software functionality frequently to their stakeholders.
  - **Managing workflow to reduce waste**
    - Understanding the current value flow, analyzing places where it is stalled, defining changes, and then implementing the changes.
  - **Making process policies explicit**
    - (e.g., write down your reasons for selecting items to work on and the criteria used to define “done”).
  - **Focusing on continuous improvement**
    - Creating feedback loops where changes are introduced based on process data and the effects of the change on the process are measured after the changes are made
  - **Make process changes collaboratively**
    - Involve all team members and other stakeholders as needed.

AGILE FRAMEWORKS: KANBAN

KANBAN BOARDS



AGILE FRAMEWORKS: XP PROGRAMMING

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- XP Planning
  - Begins with the creation of “user stories”
  - Agile team assesses each story and assigns a cost
  - Stories are grouped to for a deliverable increment
  - A commitment is made on delivery date
  - After the first increment “project velocity” is used to help define subsequent delivery dates for other increments
- XP Design
  - Follows the KIS principle
  - Encourage the use of CRC cards
  - For difficult design problems, suggests the creation of “spike solutions”—a design prototype
  - Encourages “refactoring”—an iterative refinement of the internal program design
- XP Coding
  - Recommends the construction of a unit test for a store before coding commences
  - Encourages “pair programming”
- XP Testing
  - All unit tests are executed daily
  - “Acceptance tests” are defined by the customer and executed to assess customer visible functionality

AGILE FRAMEWORKS: XP PROGRAMMING

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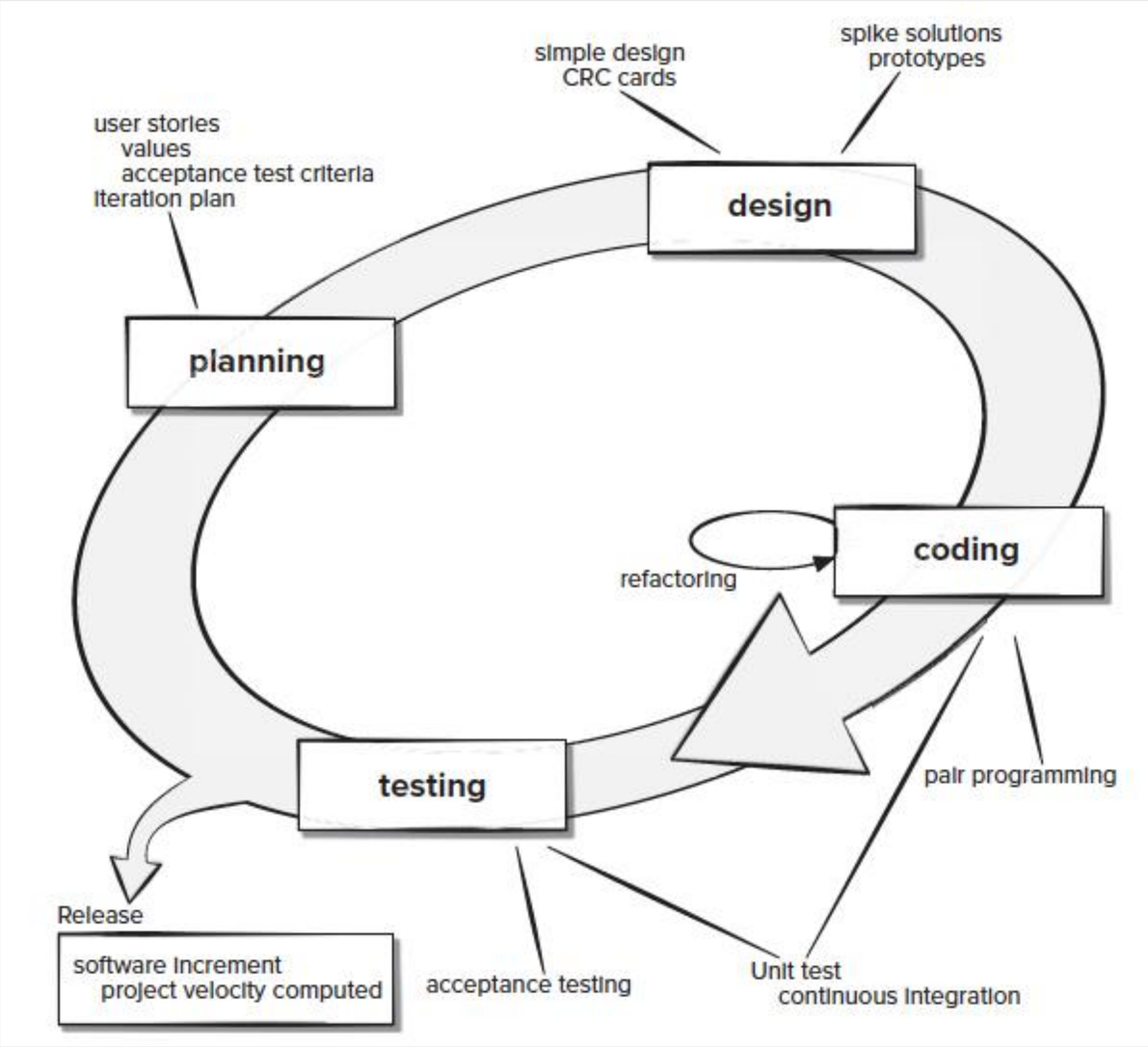

AGILE FRAMEWORKS: DEVOPS

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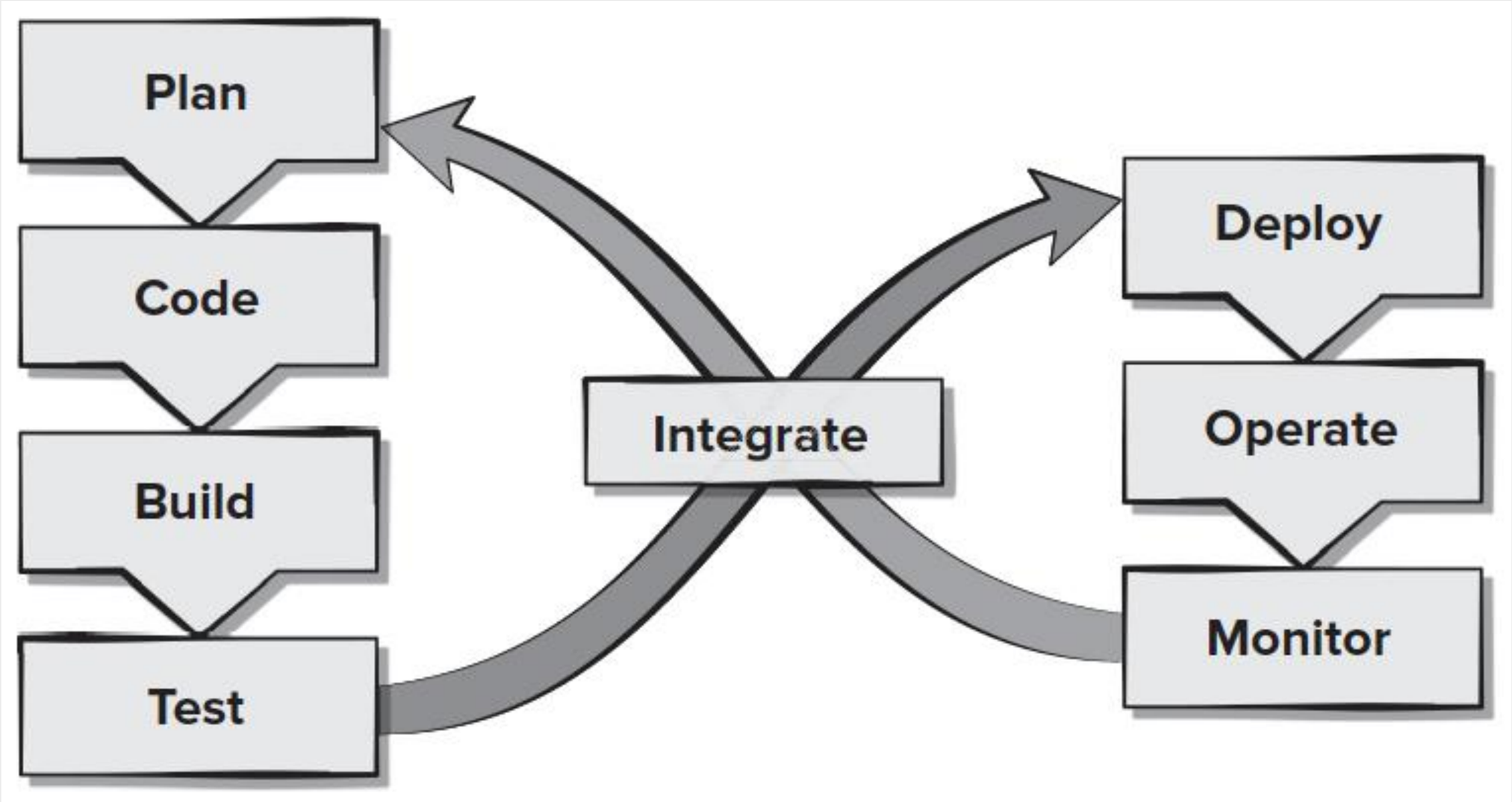
- DevOps was created by Patrick DeBois to combine Development and Operations. DevOps attempts to apply agile and lean development principles across the entire software supply chain.
- The DevOps approach involves several stages that loop continuously until the desired product exists:
  - Continuous development
    - Software deliverables are broken down and developed in multiple sprints with the increments delivered to the quality assurance14 members of the development team for testing
  - Continuous testing
    - Automated testing tools are used to help team members test multiple code increments at the same time to ensure they are free of defects prior to integration.
  - Continuous integration
    - The code pieces with new functionality are added to the existing code and to the runtime environment and then checked to ensure there are no errors after deployment.
  - Continuous deployment
    - At this stage the integrated code is deployed (installed) to the production environment, which might include multiple sites globally that need to be prepared to receive the new functionality.
  - Continuous monitoring
    - Operations staff who are members of the development team help to improve software quality by monitoring its performance in the production environment and proactively looking for possible problems before users find them.

AGILE FRAMEWORKS: DEVOPS

AGILE FRAMEWORK: DEVOPS



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